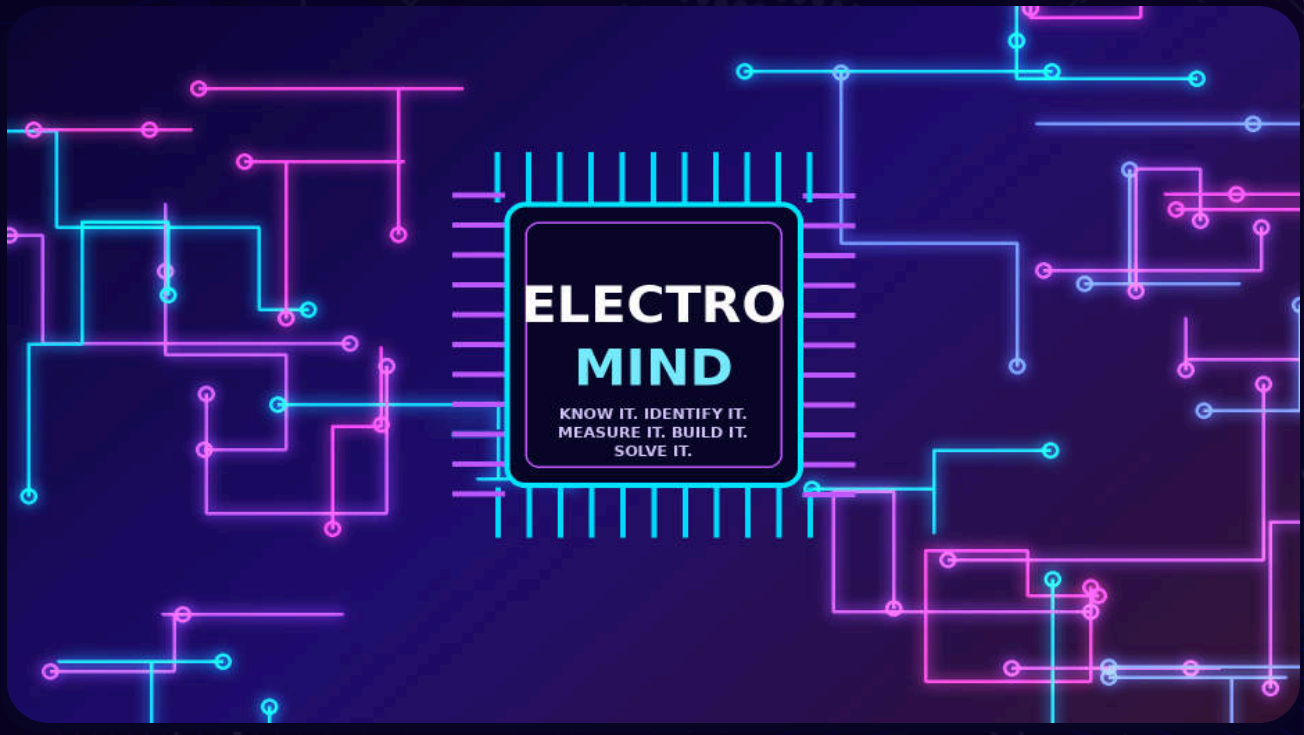
NIT ARUNACHAL PRADESH

ADDOVED

ENTER THE ARENA

ELECTROMIND



Introduction

ELECTROMIND is a three-round technical competition designed to test and develop students' technical knowledge, component identification skills, measurement ability, practical understanding, problem-solving skills, and basic circuit design capabilities.

Event Structure

ROUND 1 — ELECTROMIND QUIZ

Objective: To test the participants' fundamental knowledge of Electrical and Electronics Engineering.

Format: Participants will answer a set of multiple-choice questions covering basic and practical concepts.

Qualifying criteria for Round 2:

1. Top-performing participants.
2. Group participation (maximum allowed members is 3).
3. Time limit.

ROUND 2 — IDENTIFY & MEASURE

Objective: To evaluate participants' ability to recognize electrical/electronic components, understand their applications and perform basic measurements. Participants will be provided with real electrical and electronic components/devices.

This will be an individual practical round. Members of each team will participate one by one. Each participant will independently identify electrical/electronic components, state their applications, and perform assigned measurements.

Tasks

1. Identify the component/device.
2. State its applications.
3. Measure/determine the value.
4. Participants may also be asked to determine component values using printed markings and verify them using measuring instruments.

Judging Criteria

1. Time cost.
2. Accuracy, identification and application.
3. Total points is the sum of individual participants in a group (one by one).

Round 2 Team Score = Participant 1 + Participant 2 + Participant 3

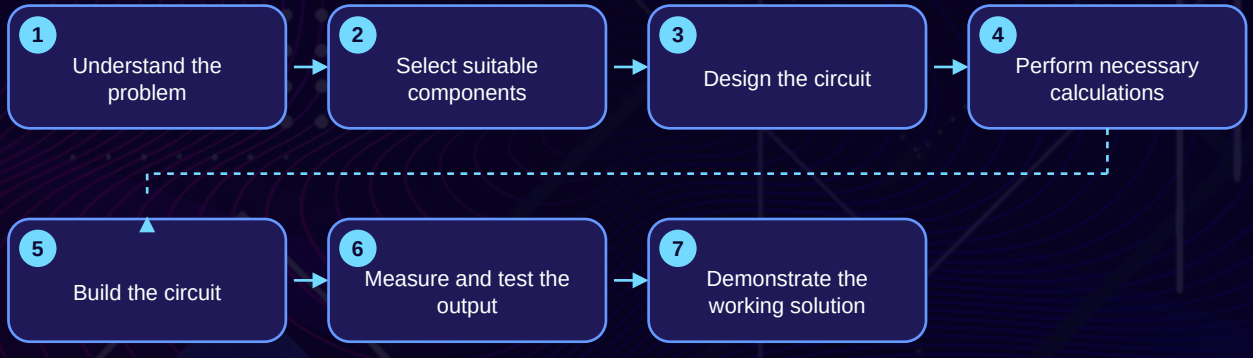
Purpose: *To test whether participants can apply theoretical knowledge to real-world components and instruments.*

ROUND 3 — BUILD THE SOLUTION

Objective: To test the participants' engineering thinking, circuit-design ability, component selection, practical implementation, measurement, and problem-solving skills.

In this round they will receive a simple real-world problem statement along with a set of available components.

How it works



Note: *Participants must decide which components are required and have to collect the components from the coordinators at the start of Round 3, in a single go. Once your turn is over, no additional components will be provided. Each team has to select only one problem statement, and it will be allocated on a first-come,*

first-served basis from the timeline of the registration portal.

Problem Statements

1. **Automatic Street Light with Adjustable Sensitivity:** Design and construct a low voltage automatic street-light circuit that switches an LED/lamp ON when surrounding light intensity falls below an adjustable threshold and switches it OFF when sufficient light is available.
2. **Temperature-Controlled Fan:** Design and construct a circuit in which a small DC fan automatically turns ON when temperature rises above an adjustable threshold and turns OFF when temperature falls below the threshold. An LED should indicate fan operation.
3. **Adjustable 3–7 V DC Power Supply:** Using a 9 V DC battery, design and construct a regulated adjustable DC power supply providing approximately 3–7 V output. Participants must calculate and verify the output voltage.
4. **Automatic Emergency Light:** Design and construct a circuit that automatically transfers the lighting load to a pre-charged backup 9 V battery when the primary 9 V supply is disconnected.
5. **Water Level Indicator with High-Level Alarm:** Design and construct a circuit that indicates LOW, MEDIUM and HIGH water levels using separate LEDs. A buzzer must activate when the water reaches HIGH level.
6. **Adjustable Timer-Based Light:** Design and construct a circuit in which pressing a push button turns an LED ON and automatically turns it OFF after an adjustable delay of approximately 5–30 seconds.
7. **Overvoltage Warning & Load Protection:** Design and construct a low-voltage load protection circuit that monitors the supply against a preset safe threshold. When the voltage exceeds the selected threshold, the circuit must disconnect the load and activate visual and audible warnings.
8. **Two-Level Battery Voltage Indicator:** Design and construct a circuit that indicates NORMAL and LOW battery conditions using green and red LEDs. When the battery voltage falls below the selected threshold, a buzzer must also activate.
9. **DC Motor Speed Control with Protection:** Design and construct a PWM-based speed-control circuit for a small low-current DC motor. Motor speed should be adjustable, and suitable protection against inductive voltage must be provided.
10. **Rain/Water Detection Alarm:** Design and construct a low-voltage circuit that detects water/rain on a conductive sensing surface and activates both visual and audible warnings. Sensitivity should be adjustable.

General Rules

- A team may register only once. In case of multiple registrations, only the first submission will be considered.
- Only undergraduate students from any field are eligible to participate.
- Each team may consist of a maximum of 3 members.
- Use of smartphones, calculators, the internet and AI is prohibited. If anyone is found using them, the whole team will be disqualified without any warning.
- Participants must carry valid student ID cards from their respective institutions.
- In case of disputes or discrepancies, the decision of the organisers and judges will be final and binding.

- In case any component gets damaged, no extra component will be provided.
- Organisers reserve the right to modify any or all rules. Any changes will be communicated to registered participants.
- Only students of NIT Arunachal Pradesh and students of institutes in the State of Arunachal Pradesh are allowed to participate.

Prize Pool

Prize Value: ₹17,000+ (including goodies)

Registration Charge

REGISTRATION FEE

₹ 100 per team

Contact Details

EVENT COORDINATOR

Saurav Kumar

Ph **+91 9798727650**

Join our WhatsApp group for all the future updates!

ELECTROMIND
WhatsApp group

SCAN TO JOIN